Efe Tarhan

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Work Permit B

Education

École Polytechnique Fédérale de Lausanne (EPFL) — *Lausanne, Switzerland*
MSc in Communication Systems — *Sep 2024 – Present*

- GPA: 5.9/6.0
- Granted with **Excellence Scholarship**
- Specialization in **Signal & Image Processing**

Ihsan Dogramaci Bilkent University — *Ankara, Turkey*
BSc in Electical and Electronics Engineering — *Sep 2020 – Jun 2024*

- 3.95/4.0 GPA
- Ranked **2nd among 180 people** in 2020 cohort.
- Specialization in **Data Science**

Experience

Graduate Research Assistant — *Lausanne, Switzerland*
École Polytechnique Fédérale de Lausanne (EPFL) — *Sep 2024 – Present*

- Developing an efficient multiresolution optimization algorithm for ptychographic imaging as part of the Biomedical Imaging Group (BIG), under the supervision of Prof. Michaël Unser.

Undergraduate Research Assistant — *Ankara, Turkey*
Bilkent University — *Mar 2023 – Sep 2024*

- Conducted research on intelligent reflecting surfaces and visible light positioning systems under the supervision of Prof. Sinan Gezici.
- Published the journal article *IRS-Aided Visible Light Positioning with a Single LED Transmitter* in **Elsevier Digital Signal Processing**.

Signal Processing Engineer — *Ankara, Turkey*
NANOTAM — *Apr 2023 – May 2024*

- Worked in the nanotechnology research facility led by Prof. Ekmel Özbay, focusing on Optical Time Domain Reflectometry (OTDR). Developed a real-time, time-efficient processing pipeline for OTDR data using statistical signal processing techniques in MATLAB, Python, and C++.

Antenna Engineer Intern — *Ankara, Turkey*
ASELSAN — *Jul 2023 – Aug 2023*

- Worked on phsysical design and material effects on tightly coupled phased array absorbers. Designed a vertically meandered Archimedean spiral antenna using Ansys HFSS software.

Computer Vision Intern — *Ankara, Turkey*
UMRAM — *June 2022 – July 2022*

- Worked on the multilabel-multiclass disease classification task of chest tomography samples. Developed deep learning based solutions using CNNs.

Publications

IRS Aided Visible Light Positioning with a Single LED Transmitter Jan 2025

Efe Tarhan, Furkan Kokdogan, Sinan Gezici

Elsevier Digital Signal Processing Journal

10.1016/j.dsp.2024.104799

Projects

A Multiresolution Framework for Ptychography *Sep 2024 – Present*

- Multiresolution Optimization, Ptychography, B-Splines, Phase Retrieval

From Pixels to Wireframes: 3D Reconstruction via CLIP-Based Sketch Abstraction *Apr 2025 – Jun 2025* *Website*

- Gaussian Splatting, Bezier Curves, Abstract Representation, CLIP

DetoxText: Text Detoxification Using Finetuned Encoder-Decoder Models *Apr 2025 – Jun 2025* *Github*

- Hate Speech Removal, Supervised Finetuning, BERT, GWEN

Large-Scale 3D Scene Relighting using Pre-Trained Diffusion Models *Mar 2025 – Jun 2025* *Github*

- NeRF, Gaussian Splatting, Diffusion Models, Scene Relighting

Road Segmentation from Aerial Images *Nov 2024 – Jan 2025* *Github*

- Semantic Segmentation, UNet, PSPNet, SegFormer

Wireless Train Signalization *Sep 2023 – Jun 2024* *Github*

- Industrial Design Project, Embedded Systems, RFID, WiFi, GPS, Cloud Database, GUI Design.

ML-Enhanced VLP in IRS-Assisted Indoor Environments with a Single LED Transmitter *Sep 2023 – Jan 2024* *Github*

- Visible Light Positioning, Intelligent Reflective Surfaces, Illumination Modelling, KNN.

Achievements & Awards

- École Polytechnique Fédérale de Lausanne **Excellence Scholarship**.
- Bilkent University Electrical and Electronics Engineering **Academical Excellence Award**.
- TÜBİTAK 2209-B Industry Oriented Projects Support Fund.
- 2247 - C STAR-Intern Researcher Scholarship.
- İhsan Doğramacı Bilkent University **Full Comprehensive Scholarship**.
- Ranked **441st among 2 million students** in university entrance exam (YKS).

Skills

Coding Languages: Python, MATLAB, C/C++, R, MySQL

Technologies: PyTorch, TensorFlow, Pandas, Google Firebase, Qt

Languages: English (Full Working Proficiency), French (Intermediate Proficiency)